

SANTIAGO RESTREPO RUIZ

Principal AI Engineer | Optimization, Cryptography & Electronics

www.waajacu.com | santiago.restrepo.ruiz@gmail.com | [LinkedIn](#) | [Oslo, Norway](#) | [From Colombia](#)

PROFILE

Software and hardware engineer focused on high-performance C/C++, robotics, circuits, cryptography, and reliable intelligent systems.

EXPERIENCE

Principal System Architect (Artificial Intelligence)

PricewaterhouseCoopers (PwC)

May 2025 – July 2026 Montevideo, Uruguay

- Leads AI system architecture for enterprise environments, aligning solution design with business strategy, security, and delivery constraints.
- Designs scalable data and application architectures for intelligent automation, analytics, and machine-learning workflows in regulated contexts.
- Translates complex technical and operational requirements into production-ready platforms, integration patterns, and implementation roadmaps.

Senior C++ Optimization Engineer

Globant

January 2023 – May 2025 Remote, Colombia

- Stabilized and optimized a large-scale C/C++ platform for construction logistics, improving reliability and performance through applied mathematics and disciplined engineering practices.
- Led development of a high-performance real-time street-mapping and routing engine designed for global-scale deployment.

Senior Data Engineer (Banking & Retail)

Fourier Analytics

March 2021 – January 2023 Remote, Colombia

- Designed and implemented data pipelines in C++ for high-volume banking and retail workloads.
- Built time-series forecasting and analytical models for operational and commercial decision-making.
- Secured APIs across teams while supporting compliance and data-protection requirements.

Lead Quantitative Analyst (Finance Engineering Lead)

AUB Capital

2020 – 2021 Medellín, Colombia

- Led analytics and engineering work around a USD 10 million private-equity portfolio.
- Developed quantitative models and automated systems for low-risk global market instruments from concept through implementation.
- Built low-latency interfaces, secure Linux infrastructure, and forecasting workflows for financial engineering operations.

RPA Developer (Financial Process Automation)

Stratesys

March 2018 – June 2020 Remote, Spain

- Delivered financial process automation (Automation Anywhere, UiPath, and Blue Prism).
- Developed supporting tooling and frameworks in Python, JavaScript, and PowerShell.
- Applied NLP and document-processing to multilingual classification and electronic-invoice extraction.

Founding Partner

Waajacu

January 2012 – Present Colombia

- Founded and led an open-source organization focused on computational diplomacy, cryptography, optimization, and high-performance software engineering.
- Researches cryptographic protocols for cooperation, markets, law, and robotics.
- Explores entropy maximization, stochastic modeling, nonlinear dynamical systems, and optimal control.

EDUCATION

Electronic Engineering

Universidad del Quindío

February 2013 – September 2018 Armenia, Colombia

Thesis: Design and Modeling of a Delta Robot.

Mathematics and Algorithms

Massachusetts Institute of Technology (OpenCourseWare)

February 2013 – September 2018 Online

Independent study in algorithms, mathematics, and computer science through MIT OpenCourseWare.



SELECTED HIGHLIGHTS

- Top SABER PRO Engineering result (2018).
- Founder of www.waajacu.com: advanced engineering and research.
- Founder of [Save the Bees and Seeds](#), an open-source software foundation.
- Thesis: [Delta robot design and modeling](#).

CORE EXPERTISE

High-Performance C/C++
Robotics & Autonomous Systems
Embedded Systems & Electronics
Optimization & Optimal Control
Mathematical Modeling & Simulation
AI / Machine Learning Systems
LLM & Agentic AI Architecture
Applied Cryptography & Computational Diplomacy
Real-Time Systems & Algorithms
Data Engineering & Distributed Systems

TECHNICAL SKILLS

CORE STACK

Linux / Unix	●●●●●
Windows	●●●●●
C / C++	●●●●●
Python / JavaScript	●●●●●
Rust / Go / PHP	●●●●●
FPGA / Verilog / VHDL	●●●●●
SQL / NoSQL	●●●●●
Git Docker / Kubernetes / Nginx	●●●●●
GDB / Valgrind	●●●●●
ADK / LangChain / LangGraph	●●●●●
PyTorch / TensorFlow	●●●●●

APPLIED DOMAINS

Mathematical Modeling	●●●●●
Entropy / Information Theory	●●●●●
Control / Geometry	●●●●●
Differential Privacy	●●●●●
Probability / Reinforcement Learning	●●●●●
Embedded Systems / Networking	●●●●●
Computational Diplomacy	●●●●●

LANGUAGES

Spanish	●●●●●
English	●●●●●
Chinese (Simplified)	●●●●●